



Republic of the Philippines

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Vice President for Academic Affairs

COLLEGE OF COMPUTER AND INFORMATION SCIENCES

Course Title : **Database Administration**

Course Code : **COMP 018**

Course Credit : 3 UNITS / 5 HOURS (2 Units Lecture And 1 Unit Lab)

Pre-Requisite :

Course Description : This course discusses basic concepts on database administration. It explores the features of technology specific database packages such as IBM DB2 UDB, Microsoft SQL Server, MySQL, Sybase, and Oracle among others. It includes discussions on installation, creation and configuration of databases, configure security, and transfer data between sources/servers, backing up and restoring databases, database concurrency, and optimizing database performance.

INSTITUTIONAL INTENDED LEARNING OUTCOMES (ILO)	PROGRAM INTENDED LEARNING OUTCOMES (PILO) COLLEGE		COURSE INTENDED LEARNING OUTCOMES (CILO) SUBJECT	
	BSIT	Course Outcomes		Course Outcomes
Creative and Critical Thinking	IT01	Apply knowledge of computing, science, and mathematics appropriate to the discipline.	LO1	Discuss the concept of data and database administrations. Explain the role of database administrator in the management of databases, pertaining to the security, availability, quality of databases
	IT03	Analyze complex problems, and identify and define the computing requirements appropriate to its solution.		
	IT04	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer based systems.	LO2	Identify specific SQL clauses that should be used in code development to be able to generate the required outputs
	IT05	Design, implement, and evaluate computer based systems, processes, components, or programs to meet desired needs and requirements under various constraints.	LO3	Evaluate existing database management platforms which will suit the current and future needs of an organization
	IT06	Integrate IT-based solutions into the user environment effectively.	LO4	Implement a robust application using a specific DBMS which has been tested against the requirements of a particular business
	IT09	Assist in the creation of an effective IT project plan.	LO5	Apply test scenarios to make sure application works as expected; Ensure seamless implementation of DBMS programs /applications for the business or organization
			LO6	Identify all specific activities in each phase of a project plan from the planning to the implementation. Incorporate into the plan the scheduling of resources such as



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				time, budget, and people which should make up the project team.
Adeptness in the Responsible Use of Technology	IT07	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession	LO7	Design an application code in the most efficient manner using a particular database management system; Apply the features of the DBMS (back-up, file transfers, security protocols, etc.) to maximize the benefits for application being developed with the end users in mind. Create optimal running programs and systems using SQL and specific database management system, through efficient coding schemes and effective table designs
Community Engagement	IT02	Understand best practices and standards and their applications.		
High Level of Leadership and Organizational Skills.	IT08	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.	LO8	
Strong Service Orientation				
Effective Communication	IT10	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.		
Sense of Nationalism and Global Responsiveness.	IT11	Analyze the local and global impact of computing information technology on individuals, organizations, and society.		
Sense of Personal and Professional Ethics	IT12	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.		
Passion to Life-Long Learning	IT13	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.		

COURSE PLAN:

Week	Topic	Learning Outcomes	Methodology	Resources	Assessment
Week 1	Fundamentals - Introductions (Faculty and Students) - Discussion of Course Syllabus including Grading System and General rules - Classroom Management	a.) Demonstrate an understanding of what the subject is all about, what will be in scope for the semester, and what students are expected to learn b.) Communicate with fellow students and teacher and begin to establish rapport c.) Identify and explain the course assessment and validation criteria , including grading system to understand how to pass the subject d.) Learn what are the do's and don'ts when inside the classroom	Orientation Introductions	Course Syllabus	NA
Week 2	Introduction to Database Administration - Data Administration - DBA Role - Data Security, Data Availability, Data Quality	a.) Discuss what a database administrator is and what his roles are b.) Explain the significance of security, availability, quality of a database c.) Review most used DBMS products in the market LO1, LO3	Lecture Recitation Homework	Powerpoint Presentation Video Clips on SQL	Quiz 1
Week 3-4	Review of Basic SQL on Single Tables - Structure of the basic SQL Statement - Column and Scalar Functions - GROUP BY/HAVING Clauses Basic SQL on Multiple Tables - Use of WHERE clause as join predicate	a.) Create program codes using basic SQL structure using single and multiple tables b.) Determine when and what aggregate functions to use c.) Determine appropriate scalar function to use to simplify a program requirement d.) Determine when to use a GROUP BY clause e.) Explain the difference of using HAVING and WHERE clause f.) Test program codes to ensure they meet the requirements	Lecture Recitation/Class Discussion Hands-on Exercises Homework	Powerpoint Presentation Video Clips on SQL Computer Lab Exercises	Part of Quiz 2

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		LO2, LO5, LO8			
Week 5	Review of Integrity Constraints Creating and Managing Databases Creating and Managing Tables	a.) Explain the use and importance of integrity constraints b.) Determine how to enforce integrity constraints in program codes c.) Create a program code that will create databases/tables following the rules of integrity constraints d.) Ensure that table design is optimal i.e. proper data type/size is assigned to fields LO8	Lecture Recitation/Class Discussion Hands-on Exercises	Powerpoint Presentation Computer Lab Exercises	Part of Quiz 2
Week 6	Data Manipulation Language - INSERT / DELETE / UPDATE	a.) Create program codes that will apply the DML commands b.) Test program codes to ensure they meet the requirements LO2, LO5, LO8	Lecture Recitation/Class Discussion Hands-on Exercises	Powerpoint Presentation Computer Lab Exercises	Part of Quiz 3 Graded LAB Exercises
Week 7-8	Advance SQL Other Clauses for Processing Multiple Tables - INNER/OUTER JOIN - UNION/INTERSECT	a.) Create complex program codes using specific commands for multiple table processing b.) Test program codes to ensure they meet the requirements LO2, LO5, LO8	Lecture Recitation/Class Discussion Hands-on Exercises	Powerpoint Presentation Computer Lab Exercises	Part of Quiz 3 Graded LAB Exercises
Week 9	MIDTERM				
Week 10	Subquery	a.) Analyze problems to determine when to use a subquery b.) Create program codes for problems requiring subquery c.) Test program codes to ensure they meet the requirements LO2, LO5, LO8	Lecture Recitation/Class Discussion Hands-on Exercises Homework	Powerpoint Presentation Computer Lab Exercises	Quiz 4 Graded LAB Exercises
Week 11-12	Creating VIEWS Creating Stored Procedures	a.) Determine when to use VIEWS and what are its advantages b.) Determine when to use Stored Procedures and what are its advantages c.) Create program codes for problems requiring subquery d.) Test program codes to ensure they meet the requirements LO2, LO5, LO8	Lecture Recitation Hands-on Exercises	Powerpoint Presentation Computer Lab Exercises	Part of Quiz 5
Week 13	Creating Triggers	a.) Determine when to use Stored Procedures and what are its advantages b.) Create program codes for problems requiring subquery c.) Test program codes to ensure they meet the requirements LO2, LO5, LO7	Lecture Hands-on Exercises	Powerpoint Presentation Computer Lab Exercises	Part of Quiz 5
Week 14	Importing and Exporting Data - From / Into Excel / Text Files - Data Transfers between Tables	a.) Explore the different ways of moving data from and into the database using different data storage types LO2	Lecture Hands-on Exercises	Powerpoint Presentation Computer Lab Exercises	Part of Quiz 6 Graded LAB Exercises
Week 15	Database Back-up	a.) Explain the different types of backups	Lecture	Powerpoint	Part of Quiz 6



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	- On-site / Off-site Backup - Planning and Designing Backup Procedure - Backup Implementation	- b.) Determine the different ways to do database back-up including the use of SQL codes - c.) Create program code for backup purpose LO1, LO7	Hands-on Exercises	Presentation Computer Lab Exercises	
Week 16	Database Security Database Concurrency	- a.) Discuss database security and how this is implemented - b.) Learn the concepts of database concurrency and know the equivalent command in DBMS to implement concurrency LO1, LO7	Lecture Recitation/Class Discussion	Powerpoint Presentation	Part of Quiz 6
Week 17	Project Presentation	LO4, LO6, LO7, LO8			
Week 18	FINALS				

COURSE ASSESSMENT & EVALUATION CRITERIA (GRADING & REQUIREMENTS)

ASSIGNMENTS / QUIZZES / HANDS-ON ACTIVITIES

MAJOR REQUIREMENTS

- MIDTERM AND/OR FINAL EXAM

DATABASE PROJECT FOR SMALL/MEDIUM BUSINESS

GRADING SYSTEM:

FIRST GRADING

= Class Standing (70%): Quizzes, Long Test, Recitation, Research Work, Assignment, Exercises, Projects, Recitation; Midterm Examination (30%)

SECOND GRADING

= Class Standing (70%): Quizzes, Long Test, Recitation, Research Work, Assignment, Exercises, Projects, Recitation; Final Examination (30%)

FINAL GRADE

= (FIRST GRADING + SECOND GRADING) / 2

BOOK/ COURSE REFERENCE:

REQUIRED READING:

1. Database Systems, Thomas Connolly and Carolyn Begg, 3rd edition, 2007

REFERENCES

1. Modern Database Management, 11th Ed (Ramesh Venkataraman, Jeffrey Hoffer, Heikki Topi)
2. Database Principles Design, Implementation, & Management Fundamentals (Carlos Coronel, Rob Moris)
3. Database Management System (Yadav, P.K.)
4. Inside Microsoft SQL Server 2008 (Ben, Itzik-Gan, et. al)
5. Modern Database Management by J.A. Hoffer, Ramesh Toppi, 11th edition, Benjamin-Cummins (Narosa), 2014.
6. Concepts of Database Management, Philip Pratt, 2012.
7. Fundamentals of Database Systems, Ramez Elmasri and Shamkant B. Navathe, 2007.

House Rules:

Aside from what is prescribed in the student handbook, the following are the professor's additional house rules:

1. The course is expected to have a minimum of four (4) quizzes. **No makeup tests will be given.**
2. Assignments and research projects/report works will be given throughout the semester. Such requirements shall be due as announced in class. Late submission shall be penalized with grade deductions (5% per day) or shall no longer be accepted, depending on the subject facilitator's discretion. Assignments and exercises are designed to assist you in understanding the materials presented in class, and to prepare you for the exams.
3. Students are required to attend classes regularly, including possible make-up classes. The student will be held liable for all topics covered and assignments made during his/her absence. The university guidelines on attendance and tardiness will be implemented.
4. Any evidence of copying or cheating during any examinations may result in a failing grade from the examination for all parties involved. Note that other university guidelines shall be used in dealing with this matter.



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8. DB2 SQL Workshop Guide, 2002 and other DBMS Guides 9. An Introduction to Database System by CJ Date, 8th edition, 2004 10. Database System Concepts by Silberchatz, 2002. Note: Extended Readings may be assigned by the professor. INSTRUCTIONAL MEDIA - Powerpoint Presentation - Video Presentations - Program Simulations	5. Students are advised to keep graded work until the semester has ended. 6. Contents of the syllabus are subject to modification with notification. 7. Cell phones, radios or other listening devices are not allowed to be used inside lecture and laboratory rooms to prevent any distractive interruption of the class activity. 8. No foods, drinks, cigarettes nor children are allowed inside the lecture and laboratory rooms. 9. Withdrawal and dropping from the subject should be done in accordance with existing university policies and guidelines regarding the matter.
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Revised and Prepared by:	Date:	Reviewed by:	Date:	Approved by:	Date
Gecilie C. Almiranez	September 17, 2025	Dr. Rhyan Molinar Head of Academic Programs			